

INT-332 Practical-5

set up a Continuous Integration (CI) pipeline using GitHub Actions that automatically builds and pushes a Docker image to Docker Hub whenever code is pushed. Create a workflow file that checks out the repository, logs in to Docker Hub using GitHub secrets, build the images from the Dockerfile, tags it as username/app-ci:latest, and pushes it. Also, briefly explain the role of environment variables and secret in the workflow.

2. Create GitHub Secrets

Open your repository on [GitHub](#)
Go to:

➤ Repository → Settings → Secrets and variables → Actions

3. Workflow File

Create this file:

➤ .github/workflows/docker-ci.yml

Add the following code:

```
name: Docker CI Pipeline

on:
  push:
    branches:
      - main

jobs:
  docker:
    runs-on: ubuntu-latest

    env:
      IMAGE_NAME: app-ci

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4
```

```
- name: Log in to Docker Hub
  uses: docker/login-action@v3
  with:
    username: ${ secrets.DOCKER_USERNAME }
    password: ${ secrets.DOCKER_PASSWORD }

- name: Build Docker Image
  run: |
    docker build -t ${ secrets.DOCKER_USERNAME }/${ env.IMAGE_NAME
}}:latest .

- name: Push Docker Image
  run: |
    docker push ${ secrets.DOCKER_USERNAME }/${ env.IMAGE_NAME }:latest
```

4. How This Workflow Works

Trigger

```
on:
  push:
    branches:
      - main
```

This starts the workflow automatically whenever code is pushed to the `main` branch.

Checkout Repository

```
uses: actions/checkout@v4
```

This downloads your repository code into the GitHub Actions runner.

Docker Login

```
uses: docker/login-action@v3
```

This logs into Docker Hub securely using GitHub Secrets.

Build Docker Image

```
docker build -t username/app-ci:latest .
```

The screenshot shows a web browser window displaying the GitHub Actions page for the repository 'rajyadav23/java-ci-maven'. The browser's address bar shows the URL 'https://github.com/rajyadav23/java-ci-maven/actions'. The page header includes the repository name and a search bar. Below the header, there are tabs for 'Code', 'Issues', 'Pull requests', 'Agents', 'Actions', 'Projects', 'Wiki', 'Security and quality', 'Insights', and 'Settings'. The 'Actions' tab is selected, showing a list of workflow runs. On the left side, there is a sidebar with the 'Actions' section, a 'New workflow' button, and a list of workflow categories: 'All workflows', 'Java CI with Maven', 'Management', 'Caches', 'Attestations', 'Runners', 'Usage metrics', and 'Performance metrics'. The main content area is titled 'All workflows' and shows '2 workflow runs'. The first run is 'Update maven.yml' with a green checkmark, indicating success. It was triggered by a push to the 'main' branch (commit 41f9647) by user 'rajyadav23' on April 10, 12:47 PM GMT+5:30, and took 21s to complete. The second run is 'Create maven.yml' with a red X, indicating failure. It was triggered by a push to the 'main' branch (commit adc26fe) by user 'rajyadav23' on April 10, 12:33 PM GMT+5:30, and took 25s to complete. The Windows taskbar at the bottom shows the system clock as 12:21 PM on 5/13/2026, along with weather information (93°F Sunny) and various application icons.

Workflow runs for **rajyadav23 / java-ci-maven**

Showing runs from all workflows

2 workflow runs

Workflow	Event	Status	Branch	Actor	Time
Update maven.yml	Push	Success	main	rajyadav23	Apr 10, 12:47 PM GMT+5:30
Create maven.yml	Push	Failure	main	rajyadav23	Apr 10, 12:33 PM GMT+5:30